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\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\SpinEchoFieldMap\_AP

TA: 32 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| TE                       | 66.00 ms          |
| Averages                 | 1                 |
| Multi-band accel. factor | 1                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 8036.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 3        |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 208 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Hamming                 | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 1                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8036.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 3        |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 3        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                          |          |
|--------------------------|----------|
| Min. prep scans          | 0        |
| Delay before PC scans    | 0 us     |
| SENSE1 coil combine      | Off      |
| Invert RO/PE polarity    | Off      |
| Force GPA balance        | Off      |
| Disable freq. update     | Off      |
| Suppress 16-bit DICOM    | On       |
| Force equal slice timing | Off      |
| FFT scale factor         | 1.00     |
| Physio recording         | Off      |
| Triggering scheme        | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\SpinEchoFieldMap\_PA

TA: 32 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| TE                       | 66.00 ms          |
| Averages                 | 1                 |
| Multi-band accel. factor | 1                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 8036.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 3        |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 208 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Hamming                 | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 1                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8036.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 3        |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 3        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                          |          |
|--------------------------|----------|
| Min. prep scans          | 0        |
| Delay before PC scans    | 0 us     |
| SENSE1 coil combine      | Off      |
| Invert RO/PE polarity    | On       |
| Force GPA balance        | Off      |
| Disable freq. update     | Off      |
| Suppress 16-bit DICOM    | On       |
| Force equal slice timing | Off      |
| FFT scale factor         | 1.00     |
| Physio recording         | Off      |
| Triggering scheme        | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\rfMRI\_REST\_AP

TA: 5:46 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 800.0 ms          |
| TE                       | 37.00 ms          |
| Averages                 | 1                 |
| Multi-band accel. factor | 8                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 800.0 ms       |
| TE                 | 37.00 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 52 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 1              |
| Reconstruction     | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 420      |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 208 mm  |
| FOV Phase | 100.0 % |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 2.0 mm |
| Base Resolution  | 104    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                         |         |
|-------------------------|---------|
| Raw Filter              | Off     |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | Off     |
| Static Field Correction | Off     |
| Normalize               | Prescan |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 800.0 ms          |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 8                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |          |
|--------------------------|----------|
| 1st Signal/Mode          | None     |
| TR                       | 800.0 ms |
| Log Signals              | Off      |
| Multi-band accel. factor | 8        |

**BOLD**

|                         |     |
|-------------------------|-----|
| GLM Statistics          | Off |
| Ignore Meas. at Start   | 0   |
| Ignore After Transition | 0   |

**BOLD**

|                         |          |
|-------------------------|----------|
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 3        |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 420      |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |          |
|--------------------------|----------|
| Excite pulse duration    | 6600 us  |
| Min. prep scans          | 0        |
| Min. prep scans SB       | 0        |
| Delay before PC scans    | 0 us     |
| Single-band images       | On       |
| MB LeakBlock kernel      | On       |
| MB dual kernel           | Off      |
| MB RF phase scramble     | Off      |
| Opt. MB RF pulse BW      | Off      |
| SENSE1 coil combine      | Off      |
| Invert RO/PE polarity    | Off      |
| Disable freq. update     | Off      |
| Suppress 16-bit DICOM    | On       |
| Force equal slice timing | Off      |
| Online multi-band recon. | Online   |
| FFT scale factor         | 1.00     |
| Physio recording         | Off      |
| Triggering scheme        | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\rfMRI\_REST\_PA

TA: 5:46 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 800.0 ms          |
| TE                       | 37.00 ms          |
| Averages                 | 1                 |
| Multi-band accel. factor | 8                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                    |                |
|--------------------|----------------|
| TR                 | 800.0 ms       |
| TE                 | 37.00 ms       |
| MTC                | Off            |
| Magn. Preparation  | None           |
| Flip Angle         | 52 deg         |
| Fat-Water Contrast | Fat Saturation |
| Contrasts          | 1              |
| Reconstruction     | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 420      |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 208 mm  |
| FOV Phase | 100.0 % |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 2.0 mm |
| Base Resolution  | 104    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                         |         |
|-------------------------|---------|
| Raw Filter              | Off     |
| Elliptical Filter       | Off     |
| Hamming                 | Off     |
| Distortion Correction   | Off     |
| Static Field Correction | Off     |
| Normalize               | Prescan |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 800.0 ms          |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 8                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|



**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Auto Coil Select |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Sum of Squares   |
| Matrix Optimization | Performance      |
| Coil Focus          | Flat             |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |          |
|--------------------------|----------|
| 1st Signal/Mode          | None     |
| TR                       | 800.0 ms |
| Log Signals              | Off      |
| Multi-band accel. factor | 8        |

**BOLD**

|                         |     |
|-------------------------|-----|
| GLM Statistics          | Off |
| Ignore Meas. at Start   | 0   |
| Ignore After Transition | 0   |

**BOLD**

|                         |          |
|-------------------------|----------|
| Model Transition States | On       |
| Temp. Highpass Filter   | On       |
| Threshold               | 4.00     |
| Paradigm Size           | 3        |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 420      |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epfid       |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Flow Compensation | None        |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
| RF Spoiling  | Off |

**Sequence - Special**

|                          |          |
|--------------------------|----------|
| Excite pulse duration    | 6600 us  |
| Min. prep scans          | 0        |
| Min. prep scans SB       | 0        |
| Delay before PC scans    | 0 us     |
| Single-band images       | On       |
| MB LeakBlock kernel      | Off      |
| MB dual kernel           | Off      |
| MB RF phase scramble     | Off      |
| Opt. MB RF pulse BW      | Off      |
| SENSE1 coil combine      | Off      |
| Invert RO/PE polarity    | On       |
| Disable freq. update     | Off      |
| Suppress 16-bit DICOM    | On       |
| Force equal slice timing | Off      |
| Online multi-band recon. | Online   |
| FFT scale factor         | 1.00     |
| Physio recording         | Off      |
| Triggering scheme        | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\SpinEchoFieldMap\_AP

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| TE                       | 66.00 ms          |
| Averages                 | 1                 |
| Multi-band accel. factor | 1                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 8036.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 1        |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 208 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Hamming                 | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 1                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8036.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | Off      |
| Temp. Highpass Filter   | Off      |
| Threshold               | 4.00     |
| Paradigm Size           | 3        |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 1        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                          |          |
|--------------------------|----------|
| Min. prep scans          | 0        |
| Delay before PC scans    | 0 us     |
| SENSE1 coil combine      | Off      |
| Invert RO/PE polarity    | Off      |
| Force GPA balance        | Off      |
| Disable freq. update     | Off      |
| Suppress 16-bit DICOM    | On       |
| Force equal slice timing | Off      |
| FFT scale factor         | 1.00     |
| Physio recording         | Off      |
| Triggering scheme        | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

## \\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\SpinEchoFieldMap\_PA

TA: 8 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| TE                       | 66.00 ms          |
| Averages                 | 1                 |
| Multi-band accel. factor | 1                 |
| AutoAlign                | Head > Brain      |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 8036.0 ms      |
| TE                    | 66.00 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 90 deg         |
| Refocus flip angle    | 180 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Disabled       |
| Contrasts             | 1              |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 1        |
| Delay in TR  | 0.00 ms  |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 208 mm  |
| FOV Phase        | 100.0 % |
| Slice Thickness  | 2.0 mm  |
| Base Resolution  | 104     |
| Phase Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | Off  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Hamming                 | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Geometry - Common**

|                          |                   |
|--------------------------|-------------------|
| Slice Group              | 1                 |
| Slices                   | 72                |
| Distance Factor          | 0 %               |
| Position                 | L0.0 P0.0 H6.0 mm |
| Orientation              | T > C-16.0        |
| Phase Encoding Dir.      | A >> P            |
| Phase Oversampling       | 0 %               |
| FOV Read                 | 208 mm            |
| FOV Phase                | 100.0 %           |
| Slice Thickness          | 2.0 mm            |
| TR                       | 8036.0 ms         |
| Multi-Slice Mode         | Interleaved       |
| Series                   | Interleaved       |
| Multi-band accel. factor | 1                 |

**Geometry - AutoAlign**

|                     |                   |
|---------------------|-------------------|
| Slice Group         | 1                 |
| Position            | L0.0 P0.0 H6.0 mm |
| Orientation         | T > C-16.0        |
| Phase Encoding Dir. | A >> P            |
| AutoAlign           | Head > Brain      |
| Initial Position    | L0.0 P0.0 H6.0    |
| L                   | 0.0 mm            |
| P                   | 0.0 mm            |
| H                   | 6.0 mm            |
| Initial Orientation | T > C             |
| T > C               | -16.00            |
| > S                 | 0.00              |
| Initial Rotation    | 0.00 deg          |

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | On       |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                   |
|-------------|-------------------|
| Position    | L0.0 P0.0 H6.0 mm |
| Orientation | T > C-16.0        |
| Rotation    | 0.00 deg          |
| A >> P      | 208 mm            |
| R >> L      | 208 mm            |
| F >> H      | 144 mm            |
| Reset       | Off               |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 8036.0 ms |
| Log Signals              | Off       |
| Multi-band accel. factor | 1         |

**BOLD**

|                         |          |
|-------------------------|----------|
| GLM Statistics          | Off      |
| Ignore Meas. at Start   | 0        |
| Ignore After Transition | 0        |
| Model Transition States | Off      |
| Temp. Highpass Filter   | Off      |
| Threshold               | 4.00     |
| Paradigm Size           | 3        |
| Meas[1]                 | Baseline |
| Meas[2]                 | Baseline |
| Meas[3]                 | Active   |
| Motion Correction       | Off      |
| Spatial Filter          | Off      |
| Measurements            | 1        |
| Delay in TR             | 0.00 ms  |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| RF Pulse Type     | Normal      |
| Gradient Mode     | Performance |
| Bandwidth         | 2290 Hz/Px  |
| Echo Spacing      | 0.58 ms     |
| Free Echo Spacing | On          |
| EPI Factor        | 104         |

**Sequence - Part 2**

|              |     |
|--------------|-----|
| Introduction | Off |
|--------------|-----|

**Sequence - Special**

|                          |          |
|--------------------------|----------|
| Min. prep scans          | 0        |
| Delay before PC scans    | 0 us     |
| SENSE1 coil combine      | Off      |
| Invert RO/PE polarity    | On       |
| Force GPA balance        | Off      |
| Disable freq. update     | Off      |
| Suppress 16-bit DICOM    | On       |
| Force equal slice timing | Off      |
| FFT scale factor         | 1.00     |
| Physio recording         | Off      |
| Triggering scheme        | Standard |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir97\_AP

TA: 6:05 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 97                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 3840 us |
| Refocus pulse duration  | 7680 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | On      |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | On      |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | On      |
| Invert RO/PE polarity   | Off     |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|



\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir97\_PA

TA: 6:05 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 97                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 2560 us |
| Refocus pulse duration  | 5120 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | Off     |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | Off     |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | Off     |
| Invert RO/PE polarity   | On      |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir98\_AP

TA: 6:08 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 98                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 3840 us |
| Refocus pulse duration  | 7680 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | On      |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | On      |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | On      |
| Invert RO/PE polarity   | Off     |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir98\_PA

TA: 6:08 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 98                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 2560 us |
| Refocus pulse duration  | 5120 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | Off     |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | Off     |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | Off     |
| Invert RO/PE polarity   | On      |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |



**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir99\_AP

TA: 6:12 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 99                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 3840 us |
| Refocus pulse duration  | 7680 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | On      |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | On      |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | On      |
| Invert RO/PE polarity   | Off     |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir99\_PA

TA: 6:12 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 99                     |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 2560 us |
| Refocus pulse duration  | 5120 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | Off     |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | Off     |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | Off     |
| Invert RO/PE polarity   | On      |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir107\_AP

TA: 6:40 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**



**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 107                    |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 3840 us |
| Refocus pulse duration  | 7680 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | On      |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | On      |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | On      |
| Invert RO/PE polarity   | Off     |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\Research\\Investigators - validated on FIT\\Frederick\\Rebuilt\_HCP\_scans\\dMRI\_dir107\_PA

TA: 6:40 min Coil Selection: Auto Voxel Size: 1.5×1.5×1.5 mm<sup>3</sup> Acc:: None Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | On                 |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Resolution - Common**

|                  |        |
|------------------|--------|
| Slice Thickness  | 1.5 mm |
| Base Resolution  | 140    |
| Phase Resolution | 100 %  |
| Interpolation    | Off    |

**Resolution - Acceleration**

|                       |      |
|-----------------------|------|
| Acceleration Mode     | None |
| Phase Partial Fourier | 6/8  |

**Resolution - Filter**

|                         |     |
|-------------------------|-----|
| Raw Filter              | Off |
| Elliptical Filter       | Off |
| Distortion Correction   | Off |
| Static Field Correction | Off |
| Normalize               | Off |

**Routine**

|                          |              |
|--------------------------|--------------|
| Slice Group              | 1            |
| Slices                   | 92           |
| Distance Factor          | 0 %          |
| Position                 | Isocenter    |
| Orientation              | T > C-16.0   |
| Phase Encoding Dir.      | A >> P       |
| Phase Oversampling       | 0 %          |
| FOV Read                 | 210 mm       |
| FOV Phase                | 100.0 %      |
| Slice Thickness          | 1.5 mm       |
| TR                       | 3526.0 ms    |
| TE                       | 89.40 ms     |
| Multi-band accel. factor | 4            |
| AutoAlign                | Head > Brain |

**Geometry - Common**

|                          |             |
|--------------------------|-------------|
| Slice Group              | 1           |
| Slices                   | 92          |
| Distance Factor          | 0 %         |
| Position                 | Isocenter   |
| Orientation              | T > C-16.0  |
| Phase Encoding Dir.      | A >> P      |
| Phase Oversampling       | 0 %         |
| FOV Read                 | 210 mm      |
| FOV Phase                | 100.0 %     |
| Slice Thickness          | 1.5 mm      |
| TR                       | 3526.0 ms   |
| Multi-Slice Mode         | Interleaved |
| Series                   | Interleaved |
| Multi-band accel. factor | 4           |

**Contrast - Common**

|                       |                |
|-----------------------|----------------|
| TR                    | 3526.0 ms      |
| TE                    | 89.40 ms       |
| MTC                   | Off            |
| Magn. Preparation     | None           |
| Flip Angle            | 78 deg         |
| Refocus flip angle    | 160 deg        |
| Fat-Water Contrast    | Fat Saturation |
| Grad. rev. fat suppr. | Enabled        |
| Reconstruction        | Magnitude      |

**Contrast - Dynamic**

|                 |          |
|-----------------|----------|
| Dynamic Mode    | Standard |
| Multiple Series | Off      |
| Delay in TR     | 0.00 ms  |

**Resolution - Common**

|           |         |
|-----------|---------|
| FOV Read  | 210 mm  |
| FOV Phase | 100.0 % |

**Geometry - AutoAlign**

|                     |                |
|---------------------|----------------|
| Slice Group         | 1              |
| Position            | Isocenter      |
| Orientation         | T > C-16.0     |
| Phase Encoding Dir. | A >> P         |
| AutoAlign           | Head > Brain   |
| Initial Position    | L0.0 P0.0 H6.0 |
| L                   | 0.0 mm         |
| P                   | 0.0 mm         |
| H                   | 6.0 mm         |
| Initial Orientation | T > C          |
| T > C               | -16.00         |
| > S                 | 0.00           |
| Initial Rotation    | 0.00 deg       |

**Geometry - Navigator**

**Geometry - Saturation**

|                    |      |
|--------------------|------|
| Special Saturation | None |
|--------------------|------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 6 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                   |
|---------------------|-------------------|
| Coil Selection      | ACS All but spine |
| Radial Sorting      | Off               |
| MSMA                | S - C - T         |
| Sagittal            | R >> L            |
| Coronal             | A >> P            |
| Transversal         | F >> H            |
| Coil Combination    | Sum of Squares    |
| Matrix Optimization | Performance       |
| Coil Focus          | Flat              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Standard |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |            |
|-------------|------------|
| Position    | Isocenter  |
| Orientation | T > C-16.0 |
| Rotation    | 0.00 deg   |
| A >> P      | 210 mm     |
| R >> L      | 210 mm     |
| F >> H      | 138 mm     |
| Reset       | Off        |

**System - pTx**

|            |          |
|------------|----------|
| B1 Shim    | TrueForm |
| Excitation | Standard |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.253118 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Image Scaling       | 1.000          |

**Physio - Signal**

|                          |           |
|--------------------------|-----------|
| 1st Signal/Mode          | None      |
| TR                       | 3526.0 ms |
| Multi-band accel. factor | 4         |

**Physio - PACE**

|               |     |
|---------------|-----|
| Resp. Control | Off |
|---------------|-----|

**Physio - PACE**

|                          |   |
|--------------------------|---|
| Multi-band accel. factor | 4 |
|--------------------------|---|

**Diff**

|                          |                        |
|--------------------------|------------------------|
| Diffusion Mode           | Free                   |
| Diff. Directions         | 107                    |
| Diffusion Scheme         | Monopolar              |
| Diff. Weightings         | 2                      |
| b-value 1                | 0 s/mm <sup>2</sup>    |
| b-value 2                | 3000 s/mm <sup>2</sup> |
| Averages 1               | 1                      |
| Averages 2               | 1                      |
| Dynamic Field Correction | Off                    |
| Invert Gray Scale        | Off                    |
| Diff. Weighted Images    | On                     |
| Trace Weighted Images    | On                     |
| Tensor                   | On                     |
| FA Maps                  | On                     |
| ADC Maps                 | On                     |
| Exponential ADC Maps     | Off                    |
| ADC Noise Threshold      | 40                     |
| Calculated Image         | Off                    |

**Sequence - Part 1**

|                   |             |
|-------------------|-------------|
| Sequence Name     | epse        |
| Dimension         | 2D          |
| Excitation        | Standard    |
| Gradient Mode     | Performance |
| Bandwidth         | 1700 Hz/Px  |
| Echo Spacing      | 0.69 ms     |
| Free Echo Spacing | Off         |
| EPI Factor        | 140         |

**Sequence - Part 2**

|                  |          |
|------------------|----------|
| Introduction     | On       |
| RF Spoiling      | Off      |
| Phase Correction | Internal |

**Sequence - Special**

|                         |         |
|-------------------------|---------|
| Excite pulse duration   | 3840 us |
| Refocus pulse duration  | 7680 us |
| Min. prep scans         | 0       |
| Min. prep scans SB      | 0       |
| Delay before PC scans   | 0 us    |
| Single-band images      | On      |
| MB LeakBlock kernel     | On      |
| MB dual kernel          | Off     |
| MB RF phase scramble    | On      |
| Opt. MB RF pulse BW     | Off     |
| Time-shifted MB RF      | Off     |
| SENSE1 coil combine     | On      |
| Invert RO/PE polarity   | On      |
| PF omits higher k-space | Off     |
| Force GPA balance       | Off     |
| Disable B1 control loop | Off     |

**Sequence - Special**

|                          |        |
|--------------------------|--------|
| Disable freq. update     | Off    |
| Suppress 16-bit DICOM    | On     |
| Force equal slice timing | Off    |
| Online multi-band recon. | Online |
| FFT scale factor         | 1.00   |
| Physio recording         | Off    |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|